

PRATIK MANDANI

PRODUCT DEVELOPMENT & FULL-STACK ENGINEER

Web Architecture (PHP / Laravel / Node.js) • Embedded Firmware & IoT (ESP32 / C/C++)

 Rajkot, Gujarat, India  pratikmandani61508@gmail.com  pratik-mandani.github.io

 linkedin.com/in/pratik-mandani  github.com/pratik-mandani

PROFESSIONAL SUMMARY

Engineer with **7 years of hands-on experience** spanning vending-machine product development, electronics, hardware integration, machine testing, R&D, and web application development. Currently working with PHP/Laravel delivering **100+ commercial websites and web platforms**, including Core Web Vitals optimization achieving **99/100 Lighthouse performance** and Oms blocking time. Actively developing embedded IoT firmware projects with ESP32 and C/C++.

TECHNICAL SKILLS

Web & Backend:	PHP, Laravel, Node.js, Express.js, MySQL, RESTful APIs, MVC Architecture, Database Design
Frontend & UI:	JavaScript (ES6+), Tailwind CSS, Bootstrap, HTML5, CSS3, Responsive Design, Figma to Code
Embedded & Firmware:	ESP32 Dual-Core, C/C++, Arduino IDE, ESP-IDF, I2C, I2S DMA, ADC, GPIO, LittleFS, State Machines
AI & Audio Pipelines:	Google Gemini (Free API Tokens), Groq LLaMA 3.3, Groq Whisper-v3 STT, Piper Neural TTS (ONNX), FFmpeg
Hardware & Electronics:	Proteus 8 CAD, PCB Assembly, Wire Harnessing, LM2596 Regulators, PC817 Optocouplers, Testing
Tools & Optimization:	Core Web Vitals, Google Lighthouse (99 Desktop / 93 Mobile), Git, GitHub, Linux, Apache

PROFESSIONAL EXPERIENCE

- Full-Stack Web Developer — [Altius Infoway](#)

May 2022 – Present (4 Years)

 - Architected and deployed **over 100 websites and web applications** across corporate, business, e-commerce, and management applications with Laravel and MySQL.
 - Directed hands-on frontend styling, custom Laravel development, database integration, maintenance, and production deployment.
 - Built role-based administrative panels, CRM lead tracking modules, and secure checkout workflows for enterprise clients.
 - Audited and tuned Core Web Vitals to achieve a verified **99/100 Desktop** and **93/100 Mobile** Lighthouse Performance score with **Oms Total Blocking Time**.
- Product Development & Electronics Engineer — [Symbian Vending Solutions](#)

Nov 2018 – Mar 2022 (3 Years)

 - Executed machine assembly, internal wiring, harness fabrication, and hardware integration for commercial beverage and cup-dispensing vending machines.
 - Conducted systematic machine testing, operational validation, issue identification, and R&D support across diverse vending prototypes.
 - Facilitated hardware-firmware coordination with development teams to resolve electrical noise and operational timing constraints.
 - Supported engineering of reusable vending control cards, 24V–12V coin interface modules, and automatic cup dispensing mechanisms.

KEY PROJECTS & HARDWARE IMPLEMENTATIONS

The Busa — Fashion Brand E-Commerce Platform

Laravel • MySQL •
JavaScript

99/100 LIGHTHOUSE • 0MS BLOCKING

- Engineered full e-commerce solution featuring product catalog, shopping cart, and order management with fluid responsiveness.
- Optimized asset pipelines and caching to achieve **99 Desktop / 93 Mobile Performance**, 100 Best Practices, and 0ms TBT.
- Live URL: thebusa.com

KOTL Assistant Robot — Desktop AI Voice Robot

ESP32 • Node.js • Free Gemini AI • Whisper •
Piper TTS

ACTIVE EMBEDDED AI

- Built real-time conversational robot connecting ESP32 edge hardware with a Node.js REST backend over Wi-Fi HTTP.
- Implemented 8kHz PCM voice capture (MAX9814 AGC Mic), multilingual Groq Whisper-v3 STT, free Google Gemini AI dialogue reasoning, and local Piper neural TTS (16kHz WAV) streamed to a MAX98357A I2S DAC amp while animating OLED RoboEyes expressions.

Diamorivo — Luxury Fine Jewelry E-Commerce

Laravel • MySQL • Tailwind CSS

- Implemented high-converting jewelry catalog and checkout workflow from Figma designs, ensuring complete mobile responsiveness.
- Live URL: diamorivo.com

Commercial Vending Machine Electronics & Subsystems

Proteus 8 CAD • Power Regulation • Relay Driver

- **Reusable Control Card:** Standardized multi-recipe dispensing across diverse tea/coffee machine models.
- **24V-12V Coin Interface Module:** LM2596 step-down regulation with PC817 optocoupler isolation to protect control lines.
- **Automatic Cup Dispensing:** Validated motor actuators, touch interfaces, and QR payment integration under load.

EDUCATION

Diploma in Electronics & Communication Engineering (EC)

GTU Certified

Government Polytechnic, Rajkot — Gujarat Technological University (GTU)